assets (not that such ad hoc querying would be an advisable form of interacting with, e.g., data lake contents!).

Another challenge is ensuring control over the duplication of data across domains. With each new business domain needing to be served with repurposed data, redundancy may emerge. This redundancy not only increases the costs of data storage and management, but also introduces lineage and versioning problems. Data consumers may analyze outdated, incomplete, or altered data and, as a result, reach misleading conclusions.

The third challenge is not specific to data mesh; however, we need to remember it when embarking on a data mesh journey. It lies in cross-domain analytics. Because business analysts and data scientists will not have access to a harmonized enterprise-wide data model, they may need additional platforms to aggregate and consolidate the various data products. Work on distributed data sources may also lead to spaghetti data pipelines; this would reduce their efficiency and reusability. Cross-domain access also challenges the data safeguarding and access controls, as many-to-many relationships require simultaneous access to multiple data products.

1.7.3 Organizational challenges

Organizational challenges in the data mesh context can be divided into three categories: scope, data practices, and culture. The first organizational challenge is the scope of such a paradigm shift. It requires coordinated efforts from multiple engaged parties from different business areas: IT, business units, and senior management.

Second, it does require a level of maturity in the handling of data by multiple parties. Centralized parties, data creators, managers, and users will need to learn to handle data with a certain level of proficiency.

Third, in most enterprises, going from *data as a by-product* to *data as a product* will be a substantial cultural shift. This shift will involve not only data engineers and IT departments, but also product, management, marketing, and sales departments, and most other departments working with data.